```python
n = int(input())
arr = list(map(int, input().split()))
key = int(input())

low = 0
high = n - 1
found = False

while low <= high:
    mid = (low + high) // 2

    if arr[mid] == key:
        print("Element found at position", mid + 1)
        found = True
        break
    elif arr[mid] < key:
        low = mid + 1
    else:
        high = mid - 1

if not found:
    print("Element not found")
```

input

```
5
10 20 30 40 50
30
Element found at position 3


...Program finished with exit code 0
Press ENTER to exit console.
```

```
n = int(input())
arr = list(map(int, input().split()))
key = int(input())

low = 0
high = n - 1
index = -1

while low <= high:
    mid = (low + high) // 2

    if arr[mid] == key:
        index = mid
        break
    elif arr[mid] < key:
        low = mid + 1
    else:
        high = mid - 1

print("Index =", index)
```

```
5
1 8 7 6 5
7
Index = 2


...Program finished with exit code 0
Press ENTER to exit console.
```

```python
n = int(input())
arr = list(map(int, input().split()))
key = int(input())

low = 0
high = n - 1
iterations = 0
found = False

while low <= high:
    iterations += 1
    mid = (low + high) // 2

    if arr[mid] == key:
        found = True
        break
    elif arr[mid] < key:
        low = mid + 1
    else:
        high = mid - 1

if found:
    print("Element found")
else:
    print("Element not found")

print("Iterations =", iterations)
```

input

```
5
2 4 5 8 7
7
Element not found
Iterations = 2


...Program finished with exit code 0
```

```python
n = int(input())
arr = list(map(int, input().split()))
key = int(input())

low = 0
high = n - 1
result = -1

while low <= high:
    mid = (low + high) // 2

    if arr[mid] == key:
        result = mid
        high = mid - 1
    elif arr[mid] < key:
        low = mid + 1
    else:
        high = mid - 1

if result != -1:
    print("First occurrence at index", result)
else:
    print("Element not found")
```

input

```
7
1 2 2 2 3 4 5
2
First occurrence at index 1


...Program finished with exit code 0
Press ENTER to exit console.
```

```python
n = int(input())
arr = list(map(int, input().split()))
key = int(input())

low = 0
high = n - 1
result = -1

while low <= high:
    mid = (low + high) // 2

    if arr[mid] == key:
        result = mid
        low = mid + 1
    elif arr[mid] < key:
        low = mid + 1
    else:
        high = mid - 1

if result != -1:
    print("Last occurrence at index", result)
else:
    print("Element not found")
```

```
7
1 2 2 2 2 4 5
2
Last occurrence at index 4


...Program finished with exit code 0
```

```python
def find_min_max(arr, low, high):
    if low == high:
        return arr[low], arr[low]

    mid = (low + high) // 2

    min1, max1 = find_min_max(arr, low, mid)
    min2, max2 = find_min_max(arr, mid + 1, high)

    return min(min1, min2), max(max1, max2)


n = int(input())
arr = list(map(int, input().split()))

minimum, maximum = find_min_max(arr, 0, n - 1)

print("Minimum Temperature =", minimum)
print("Maximum Temperature =", maximum)
```

```
7
25 35 26 75 85 98 15
Minimum Temperature = 15
Maximum Temperature = 98


...Program finished with exit code 0
Press ENTER to exit console.
```

```python
def find_min_max(arr, low, high):
    if low == high:
        return arr[low], arr[low]

    mid = (low + high) // 2

    min1, max1 = find_min_max(arr, low, mid)
    min2, max2 = find_min_max(arr, mid + 1, high)

    return min(min1, min2), max(max1, max2)


n = int(input())
arr = list(map(int, input().split()))

minimum, maximum = find_min_max(arr, 0, n - 1)

print("Minimum Mark =", minimum)
print("Maximum Mark =", maximum)
```

```
6
45 75 84 94 65 15
Minimum Mark = 15
Maximum Mark = 94


...Program finished with exit code 0
```

```python
def find_min_max(arr, low, high):
    if low == high:
        return arr[low], arr[low]

    mid = (low + high) // 2

    min1, max1 = find_min_max(arr, low, mid)
    min2, max2 = find_min_max(arr, mid + 1, high)

    return min(min1, min2), max(max1, max2)


n = int(input())
arr = list(map(int, input().split()))

minimum, maximum = find_min_max(arr, 0, n - 1)

print("Minimum Stock Price =", minimum)
print("Maximum Stock Price =", maximum)
```

```
7
154 124 214 356 286 784 985
Minimum Stock Price = 124
Maximum Stock Price = 985


...Program finished with exit code 0
```

```python
def find_min_max(arr, low, high):
    if low == high:
        return arr[low], arr[low]

    mid = (low + high) // 2

    min1, max1 = find_min_max(arr, low, mid)
    min2, max2 = find_min_max(arr, mid + 1, high)

    return min(min1, min2), max(max1, max2)


n = int(input())
arr = list(map(int, input().split()))

minimum, maximum = find_min_max(arr, 0, n - 1)

print("Fastest Time =", minimum)
print("Slowest Time =", maximum)
```

input

```
7
15 45 35 62 56 48 75
Fastest Time = 15
Slowest Time = 75


...Program finished with exit code 0
Press ENTER to exit console.
```

```python
def find_min_max(arr, low, high):
    if low == high:
        return arr[low], arr[low]

    mid = (low + high) // 2

    min1, max1 = find_min_max(arr, low, mid)
    min2, max2 = find_min_max(arr, mid + 1, high)

    return min(min1, min2), max(max1, max2)


n = int(input())
arr = list(map(int, input().split()))

minimum, maximum = find_min_max(arr, 0, n - 1)

print("Minimum Height =", minimum)
print("Maximum Height =", maximum)
```

```
8
457 657 257 148 369 852 147 456
Minimum Height = 147
Maximum Height = 852


...Program finished with exit code 0
Press ENTER to exit console.
```